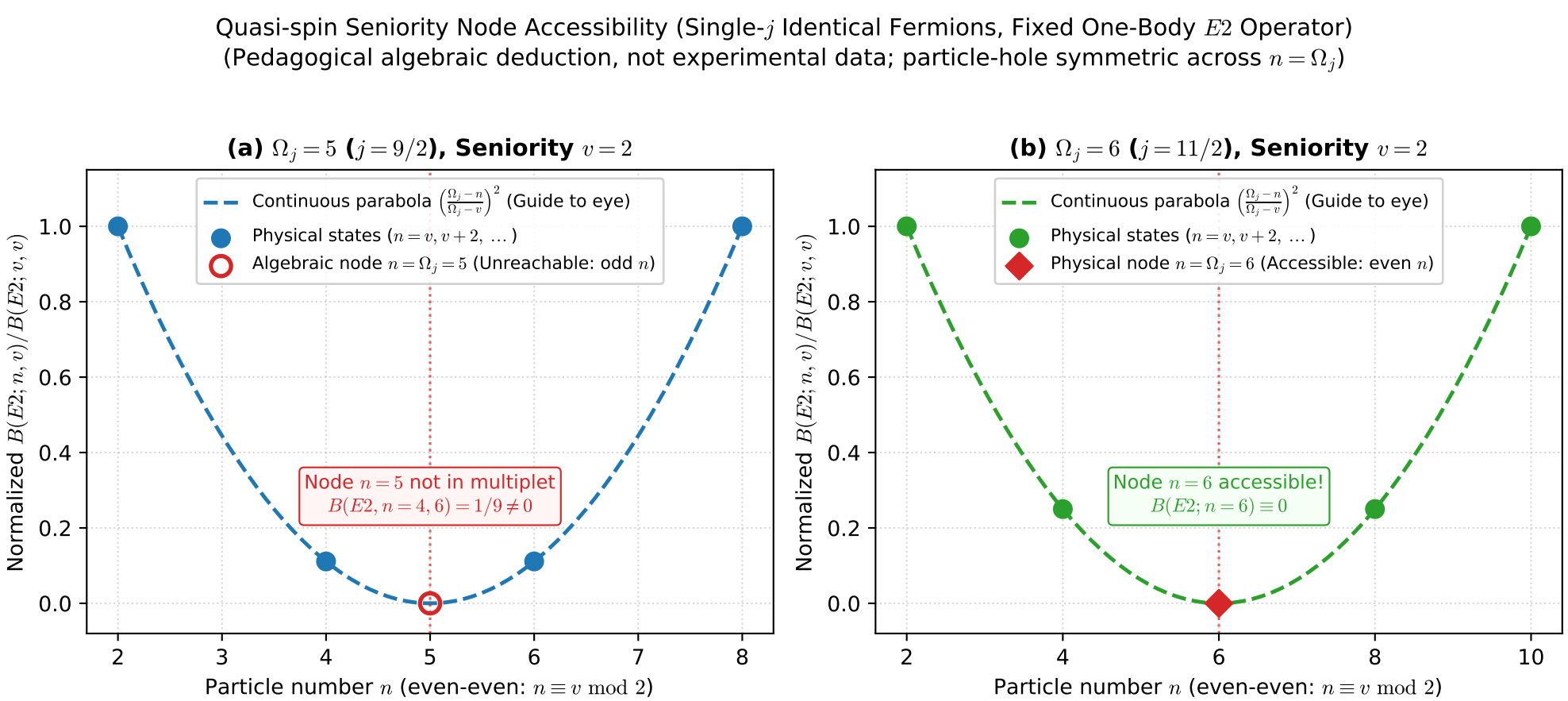
Quasi-spin Seniority Node Accessibility (Single-$j$ Identical Fermions, Fixed One-Body $E2$ Operator)
(Pedagogical algebraic deduction, not experimental data; particle-hole symmetric across $n = \Omega_j$)
(a) $\Omega_j = 5$ ($j = 9/2$), Seniority $v = 2$
Continuous parabola $\left(\frac{\Omega_j - n}{\Omega_j - v}\right)^2$ (Guide to eye)
Physical states ($n = v, v+2, \ldots$)
Algebraic node $n = \Omega_j = 5$ (Unreachable: odd $n$)
Node $n = 5$ not in multiplet
$B(E2, n = 4, 6) = 1/9 \neq 0$
Normalized $B(E2; n, v)/B(E2; v, v)$
Particle number $n$ (even-even: $n \equiv v \bmod 2$)
(b) $\Omega_j = 6$ ($j = 11/2$), Seniority $v = 2$
Continuous parabola $\left(\frac{\Omega_j - n}{\Omega_j - v}\right)^2$ (Guide to eye)
Physical states ($n = v, v+2, \ldots$)
Physical node $n = \Omega_j = 6$ (Accessible: even $n$)
Node $n = 6$ accessible!
$B(E2; n = 6) \equiv 0$
Normalized $B(E2; n, v)/B(E2; v, v)$
Particle number $n$ (even-even: $n \equiv v \bmod 2$)